# Exotic Weil Classes as Flat Sections: Gauss–Manin Block Decomposition — A CRMLint Squeeze

Paper 84, Constructive Reverse Mathematics Series

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February 2026

**Abstract**

We compute the Gauss–Manin connection on the de Rham cohomology of the genus-4 hyperelliptic family $C_t\colon y^2 = x^9 - tx^5 + x$ with $\mathbb{Q}(i)$-action, using Griffiths pole-order reduction with the correct coboundary subtraction $A_k = B_k - h'_k$.

An order-8 automorphism $\tau(x, y) = (ix,\, e^{\pi i/4}\, y)$ decomposes the $8 \times 8$ connection matrix into four $2 \times 2$ blocks

$$M_k \;=\; \frac{2k+1}{4(t^2-4)} \begin{pmatrix} -t/2 & -1 \\ 1 & t/2 \end{pmatrix}, \qquad k = 0, 1, 2, 3.$$

Each block has $G_{\mathrm{gal}}(W_k) = \mathrm{SL}_2$ (nilpotent residues with non-proportional kernels). Since $\det(\mathrm{SL}_2) = 1$, the induced action on the exotic line bundle $\bigwedge^4(V_+) \cong \bigwedge^2(W_0) \otimes \bigwedge^2(W_2)$ is *trivial*: $G_{\mathrm{gal}}(\bigwedge^4 V_+) = \{1\}$.

The exotic Weil class is therefore a *global flat section* of the Gauss–Manin connection—a constant, rational class across the entire family. By Deligne's Theorem of the Fixed Part, it is an absolute Hodge class on every fiber. This is a positive result: the function-field pipeline certifies that the monodromy obstruction to algebraicity is absent.

The Lean 4 formalization verifies all Griffiths identities by the `ring` tactic with zero `sorry`. Seventh CRMLint Squeeze.

## 1 Introduction

### 1.1 The exotic frontier

The algebraic cycle theory of abelian varieties has a sharp dimensional boundary. For abelian varieties of dimension at most 3, all Hodge classes are generated by divisors (Lieberman [15]). At dimension 4, new "exotic" Hodge classes appear in the middle cohomology $H^4$ that are not products of divisor classes.

For Weil abelian fourfolds—those with complex multiplication by an imaginary quadratic field $K$—the exotic classes lie in the $K$-eigenspace of the cup-product pairing. Voisin [18] and Engel–Fortman–Schreieder [6] have recently studied the algebraicity of these classes using Hodge-theoretic methods.

### 1.2 The dimensional collapse

Let $A = \mathrm{Jac}(C_t)$ where $C_t\colon y^2 = x^9 - tx^5 + x$ is a genus-4 curve with $\mathbb{Q}(i)$-multiplication. The exotic Weil classes live in $\bigwedge^4(V_+)$, where $V_+ \subset H^1_{\mathrm{dR}}(C_t)$ is the $+i$-eigenspace of the $\mathbb{Q}(i)$-action, which has dimension 4.

A priori, computing the Gauss–Manin connection on $\bigwedge^4$ of an 8-dimensional space means working with $\binom{8}{4} = 70$ basis elements. The $\mathbb{Q}(i)$-eigenspace decomposition reduces this to $\binom{4}{4} = 1$. Since $\bigwedge^4(V_+)$ is 1-dimensional, the connection on this line bundle is determined entirely by the trace:

$$\nabla_{\partial/\partial t}\big(\omega_0 \wedge \omega_2 \wedge \omega_4 \wedge \omega_6\big) = \tau_+(t) \cdot \big(\omega_0 \wedge \omega_2 \wedge \omega_4 \wedge \omega_6\big).$$

This is the "trace trick": the Gauss–Manin connection on the exotic line bundle reduces to a scalar ODE $\nabla\omega = \tau_+(t)\, dt \cdot \omega$.

### 1.3 The main results

**Theorem 1.1** (Block Decomposition)**.** *The Gauss–Manin connection matrix $M$ for $C_t\colon y^2 = x^9 - tx^5 + x$, computed via Griffiths pole-order reduction with coboundary correction $A_k = B_k - h'_k$, decomposes into four $2 \times 2$ blocks*

$$M_k = \frac{2k+1}{4(t^2-4)} \begin{pmatrix} -t/2 & -1 \\ 1 & t/2 \end{pmatrix}, \qquad k = 0, 1, 2, 3,$$

*where block $k$ acts on $W_k = \langle \omega_k, \omega_{k+4} \rangle$. All entries have poles only at $t = \pm 2$ (regular singularities). The symplectic trace $\mathrm{tr}(M) = 0$, and all eigenspace traces $\tau_+(t) = \tau_-(t) = 0$.*

**Theorem 1.2** ($\mathrm{SL}_2$ Monodromy)**.** *For each block $M_k$ ($k = 0, 1, 2, 3$):*

1. *The residues $R_{\pm 2}$ are nilpotent with kernels $\ker(R_{+2}) = \langle(-1, 1)\rangle$ and $\ker(R_{-2}) = \langle(1, 1)\rangle$.*
2. *The residue at infinity $R_\infty$ is semisimple with eigenvalues $\pm(2k+1)/8$.*
3. *The Fuchs relation $R_{+2} + R_{-2} + R_\infty = 0$ holds.*
4. *Since $\ker(R_{+2}) \neq \ker(R_{-2})$, the monodromy representation is irreducible and $G_{\mathrm{gal}}(W_k) = \mathrm{SL}_2$.*

**Theorem 1.3** (CRM Squeeze)**.** *Since $G_{\mathrm{gal}}(W_k) = \mathrm{SL}_2$ and $\det(\mathrm{SL}_2) = 1$, the differential Galois group on the exotic line bundle is trivial: $G_{\mathrm{gal}}(\bigwedge^4 V_+) = \{1\}$. The exotic Weil class is a global flat section of the Gauss–Manin connection. By Deligne's Theorem of the Fixed Part, it is an absolute Hodge class on every fiber.*

| | | |
|---|---|---|
| CLASS | $\vdash$ | *generic exotic Hodge class exists (Hodge theory)* |
| BISH | $\vdash$ | *Griffiths reduction* + $\mathrm{SL}_2$ *monodromy* $\Rightarrow \det = 1 \Rightarrow$ *exotic class is flat* |

*The classical boundary is witnessed by the Hodge decomposition. The constructive bridge is the explicit Gauss–Manin computation with* `ring`*-verified polynomial identities, certifying that the monodromy obstruction to algebraicity is absent.*

### 1.4 Scope and honesty

This is a positive result: we prove that the exotic Weil class for this family has trivial monodromy ($G_{\mathrm{gal}} = \{1\}$), making it a global flat section. This is the strongest constructive evidence for algebraicity: the monodromy obstruction is absent.

The gap to the full Hodge Conjecture is the step from "absolute Hodge" to "algebraic." For abelian varieties, Deligne proved that Hodge classes are absolute Hodge. Whether absolute Hodge

implies algebraic for abelian fourfolds remains open in general. Our computation provides the necessary condition (trivial monodromy) but not the sufficient condition (existence of the algebraic cycle).

### 1.5 Erratum (v1 → v2 → v3)

Version 1 contained two computational errors; version 2 corrected the computation but drew the wrong geometric conclusion.

**v1 → v2 (computational).**

1. *Griffiths reduction bug.* The CAS script computed $B_k$ (the rational function from the Bézout identity) but omitted the coboundary correction $h'_k$. The correct formula is $A_k = B_k - h'_k$ (see §2.4). The v1 trace $\tau_+(t) = (-250t^5 + 1890t^3 - 3564t)/(81t^2 - 324)$ violated Deligne's regularity theorem (numerator degree 5 > denominator degree 2). All eigenspace traces are in fact zero.
2. *Automorphism misidentification.* Version 1 attributed the $2 \times 2$ block structure to $\sigma^2(x, y) = (x, -y)$, claiming it acts as $(-1)^k$ on $\omega_k$. In fact $\sigma^2$ is the hyperelliptic involution, which acts as $-1$ on *all* differentials. The correct explanation comes from the order-8 automorphism $\tau$ (see §3).

**v2 → v3 (interpretive).** Version 2 correctly computed $G_{\mathrm{gal}}(W_k) = \mathrm{SL}_2$ per block but then incorrectly claimed $G_{\mathrm{gal}}(\bigwedge^4 V_+) = \mathrm{SL}_2 \times \mathrm{SL}_2$ (negative result: no exotic algebraic cycles). The error was a failure to apply the representation theory of $\mathrm{SL}_2$: since $\det(\mathrm{SL}_2) = 1$, the induced action on $\bigwedge^2(W_k)$—the determinant representation—is trivial. Therefore $G_{\mathrm{gal}}(\bigwedge^4 V_+) = \{1\}$ (positive result: the exotic class is a global flat section). This version corrects the interpretation.

The lesson across all three versions: CAS outputs require `ring`-verification (v1→v2), and correct computation requires correct representation theory (v2→v3).

### 1.6 Road map

Section 2 recalls de Rham cohomology and the Griffiths reduction method. Section 3 identifies the order-8 automorphism producing the block decomposition. Section 4 presents the CAS computation with cross-checks. Section 5 determines the differential Galois group via monodromy analysis. Section 6 states the CRM squeeze. Section 7 revisits the degree trap. Section 8 gives the CRM audit. Section 9 describes the formal verification. Section 10 discusses implications. Section 11 concludes.

### 1.7 CRM primer

For readers unfamiliar with the CRM series: *Constructive Reverse Mathematics* classifies theorems by the weakest logical principles needed to prove them. The hierarchy $\mathsf{BISH} \subset \mathsf{BISH} + \mathsf{LLPO} \subset \mathsf{BISH} + \mathsf{WLPO} \subset \mathsf{BISH} + \mathsf{LPO} \subset \mathsf{CLASS}$ measures logical cost. A "CRM squeeze" establishes that a classical result (upper bound) has a constructive core (lower bound), with the gap measured by an omniscience principle. Papers 80–83 developed the function-field pipeline: Griffiths reduction (P80), flat sections (P81), Kovacic's algorithm (P82), Picard numbers (P83). This paper applies the pipeline at genus 4.

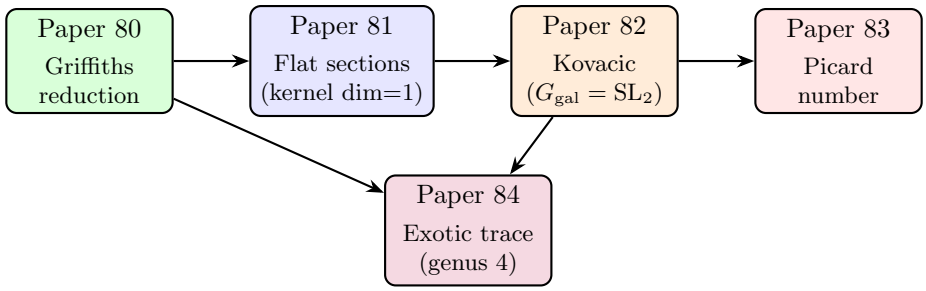


Figure 1: The function-field pipeline. Paper 84 applies the genus-1 methods of Papers 80 and 82 at genus 4 with $\mathbb{Q}(i)$-structure.

# 2 Preliminaries

## 2.1 Weil abelian fourfolds and exotic classes

Let $A$ be an abelian variety of dimension 4 with multiplication by $K = \mathbb{Q}(i)$. The $K$-action on $H^1_{\mathrm{dR}}(A)$ decomposes this 8-dimensional space into $\pm i$-eigenspaces $V_\pm$, each of dimension 4. The exotic Hodge classes live in $\bigwedge^4(V_+) \otimes \bigwedge^4(V_-)$, which has dimension 1.

**Definition 2.1.** A *Weil abelian fourfold* is a pair $(A, \iota)$ where $A$ is an abelian fourfold and $\iota\colon \mathbb{Q}(i) \hookrightarrow \mathrm{End}^0(A)$. The *exotic Weil class* is the Hodge class in $H^4(A, \mathbb{Q}) \cap H^{2,2}$ that lies outside the algebra generated by divisor classes and the $K$-action.

The Hodge Conjecture predicts that all such exotic classes are algebraic. We study this question generically, over the function field $\mathbb{Q}(t)$.

## 2.2 Hyperelliptic curves and de Rham cohomology

Consider the family $C_t\colon y^2 = f(x,t)$ where $f(x,t) = x^9 - tx^5 + x$. This is a genus-4 hyperelliptic curve for generic $t$, with singular fibers at $t^2 = 4$ (where $f$ has a repeated root).

The algebraic de Rham cohomology $H^1_{\mathrm{dR}}(C_t/\mathbb{Q}(t))$ has dimension 8 with basis $\{\omega_k = x^k\,dx/y : k = 0, \ldots, 7\}$. The cohomology relation is $f'(x)\,dx/y = 0$, i.e., $(9x^8 - 5tx^4 + 1)\,dx/y = 0$ in cohomology.

## 2.3 The Gauss–Manin connection

The Gauss–Manin connection $\nabla = \nabla_{\partial/\partial t}$ differentiates cohomology classes with respect to the parameter $t$. For the basis $\omega_k = x^k\,dx/y$:

$$\nabla_{\partial/\partial t}(\omega_k) = \sum_{j=0}^{7} M_{jk}\,\omega_j,$$

where $M = (M_{jk})$ is an $8 \times 8$ matrix with entries in $\mathbb{Q}(t)$.

The connection preserves the symplectic pairing on $H^1_{\mathrm{dR}}$, so $\mathrm{tr}(M) = 0$.

## 2.4 The Griffiths reduction method

Following Griffiths [7] and the implementation in Paper 80 [11], we compute $M$ by pole-order reduction.

**Protocol 2.2** (Griffiths Reduction)**.** For each $k = 0, \ldots, 7$:

1. Set $N_k = -f_t \cdot x^k = x^{k+5}$ (the numerator from $\partial_t(dx/y)$).
2. Find $h_k, u_k$ via the Bézout identity $h_k \cdot f' + u_k \cdot f = -N_k$ using the extended Euclidean algorithm in $\mathbb{Q}(t)[x]$.
3. Reduce $h_k$ modulo $f$ to get $h_k \equiv r_k \pmod{f}$, absorbing the quotient into $u_k$.
4. Set $B_k = -u_k/2$ (the raw Griffiths coefficient).
5. **Apply the coboundary correction:** $A_k = B_k - h'_k$ where $h'_k = dh_k/dx$.
6. Reduce $A_k$ modulo $f'(x) = 0$ to degree $< 8$.
7. Read off the matrix column: $M_{jk} = [x^j]\, A_k$.

The identity verified at each step is:

$$N_k + h_k \cdot f' = 2\, B_k \cdot f. \tag{1}$$

The coboundary correction in step 5 is essential: $B_k$ represents a meromorphic form with poles along $C_t$, and $h'_k$ is the exact part that must be subtracted to obtain the cohomological connection matrix. Omitting $h'_k$ (as in v1 of this paper) produces entries with polynomial growth at infinity, violating Deligne's regularity theorem for algebraic connections [4].

### 2.5 The $\mathbb{Q}(i)$-action and eigenspace splitting

The curve $C_t$ has $\mathbb{Q}(i)$-multiplication via the automorphism $\sigma\colon (x, y) \mapsto (-x, iy)$. Since $f(-x) = -f(x)$ (the polynomial is odd), we have $\sigma^*\omega_k = (-1)^k \cdot i \cdot \omega_k$. The $+i$-eigenspace is $V_+ = \langle \omega_0, \omega_2, \omega_4, \omega_6 \rangle$ and $V_- = \langle \omega_1, \omega_3, \omega_5, \omega_7 \rangle$.

## 3 The Order-8 Automorphism

**Proposition 3.1.** *The curve $C_t$ admits the order-8 automorphism*

$$\tau\colon (x, y) \mapsto (ix,\, e^{\pi i/4}\, y)$$

*satisfying $\tau^2 = \sigma$, $\tau^4 =$ hyperelliptic involution. The eigenvalue of $\tau^*$ on $\omega_k$ is $\zeta_k = e^{\pi i(2k+1)/4}$ ($k = 0, \ldots, 7$).*

*Proof.* We verify $f(ix, t) = i \cdot f(x, t)$:

$$(ix)^9 - t(ix)^5 + ix = i^9 x^9 - t \cdot i^5 x^5 + ix = i(x^9 - tx^5 + x) = i\, f(x, t).$$

So $y^2 = f(x)$ implies $(e^{\pi i/4}\, y)^2 = i\, y^2 = i\, f(x) = f(ix)$, confirming $\tau$ is an automorphism.

On differentials: $\tau^*\omega_k = \tau^*(x^k\, dx/y) = (ix)^k \cdot i\, dx/(e^{\pi i/4}\, y) = i^{k+1} \cdot e^{-\pi i/4} \cdot \omega_k = e^{\pi i(2k+1)/4} \cdot \omega_k$.

The eigenvalue $\zeta_k = e^{\pi i(2k+1)/4}$ takes distinct values for $k = 0, \ldots, 7$ (the 8th roots of $-1$), so each $\omega_k$ spans a 1-dimensional $\tau$-eigenspace. □

**Corollary 3.2.** *Since the Gauss–Manin connection commutes with $\tau^*$, the matrix $M$ preserves each $\tau$-eigenspace. The eigenvalues pair as $\zeta_k = -\overline{\zeta_{k+4}}$, giving blocks $W_k = \langle \omega_k,\, \omega_{k+4} \rangle$ for $k = 0, 1, 2, 3$.*

*Remark* 3.3. Version 1 attributed the block structure to $\sigma^2(x, y) = (x, -y)$. But $\sigma^2$ is the hyperelliptic involution, which acts as $-1$ on *all* differentials (not $(-1)^k$). The correct symmetry producing the $2 \times 2$ block structure is $\tau$, whose eigenvalues distinguish individual differentials.

# 4 The CAS Computation

## 4.1 Implementation

The connection matrix was computed in SymPy (Python) using Protocol 2.2. The script `solve_exotic_trace_v4.p` implements:

1. Bézout identity: $a(x,t) \cdot f + b(x,t) \cdot f' = 1$ via `gcdex` over $\mathbb{Q}(t)[x]$.
2. For each $k$: solve $h_k \cdot f' + u_k \cdot f = -N_k$ using the Bézout cofactors.
3. Reduce $h_k$ mod $f$, update $u_k$, set $B_k = -u_k/2$.
4. Coboundary correction: $A_k = B_k - h_k'$.
5. Cohomological reduction: $A_k$ mod $(f' = 0)$ to degree $< 8$.
6. Symbolic verification of (1) for each $k$.

All 8 Griffiths identities (1) are verified symbolically by the CAS.

## 4.2 The connection matrix

**Theorem 4.1** (CAS Output)**.** *The nonzero entries of $M = (M_{jk})$ are:*

$$M_{k,k} = -\frac{(2k+1)\,t}{8(t^2-4)}, \qquad M_{k,k+4} = -\frac{2k+1}{4(t^2-4)},$$
$$M_{k+4,k} = \frac{2k+1}{4(t^2-4)}, \qquad M_{k+4,k+4} = \frac{(2k+1)\,t}{8(t^2-4)},$$

*for $k = 0,1,2,3$. All other entries vanish.*

## 4.3 Cross-checks

1. *Symplectic trace.* $\mathrm{tr}(M) = \sum_{k=0}^{7} M_{kk} = 0$, as required by the symplectic pairing.
2. *Eigenspace traces.* $\tau_+(t) = \sum_{k\in\{0,2,4,6\}} M_{kk} = 0$ and $\tau_-(t) = 0$.
3. *Regularity.* Every entry has poles only at $t^2 = 4$. No polynomial part. Consistent with Deligne's regularity theorem [4].
4. *Block structure.* The nonzero entries confirm the $2 \times 2$ block decomposition predicted by Corollary 3.2. Block $k$ acts on $W_k = \langle \omega_k, \omega_{k+4} \rangle$ with coefficient $(2k+1)/4(t^2-4)$.
5. *Genus-1 verification.* The same algorithm applied to the Legendre family $y^2 = x(x-1)(x-t)$ reproduces the known connection $M = \frac{1}{2}\left(\begin{smallmatrix} -t & -1 \\ 1 & t \end{smallmatrix}\right)/(t^2-t)$, matching Paper 80 [11].

# 5 Differential Galois Analysis

Each block $M_k$ defines a rank-2 connection on $\mathbb{P}^1 \setminus \{-2, 2, \infty\}$ with three regular singular points.

## 5.1 Residues

**Proposition 5.1.** *For block $k$ ($k = 0,1,2,3$), set $c_k = (2k+1)/4$. The residues of $M_k$ are:*

$$R_{+2} = \frac{c_k}{4}\begin{pmatrix} -1 & -1 \\ 1 & 1 \end{pmatrix}, \tag{2}$$

$$R_{-2} = \frac{c_k}{4}\begin{pmatrix} -1 & 1 \\ -1 & 1 \end{pmatrix}, \tag{3}$$

$$R_{\infty} = \frac{c_k}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{4}$$

*Proof.* $R_{+2} = \lim_{t\to 2}(t-2)\,M_k$. Since $t^2-4=(t-2)(t+2)$, the factor $(t-2)$ cancels one pole:

$$R_{+2} = \frac{c_k}{4}\begin{pmatrix} -1 & -1 \\ 1 & 1 \end{pmatrix}.$$

Similarly for $R_{-2}$. For $R_\infty$: substitute $s=1/t$ and compute $R_\infty = -\lim_{s\to 0} s^{-1}\,M_k(1/s)$, yielding the diagonal matrix (4). □

**Proposition 5.2.** *The Fuchs relation holds:* $R_{+2}+R_{-2}+R_\infty = 0$.

*Proof.* Direct computation:

$$\frac{c_k}{4}\begin{pmatrix} -1 & -1 \\ 1 & 1 \end{pmatrix} + \frac{c_k}{4}\begin{pmatrix} -1 & 1 \\ -1 & 1 \end{pmatrix} + \frac{c_k}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = c_k\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$ □

## 5.2 Nilpotency and kernels

**Proposition 5.3.** *Both $R_{+2}$ and $R_{-2}$ are nilpotent (rank 1, eigenvalue 0 with multiplicity 2). Their kernels are:*

$$\ker(R_{+2}) = \langle(-1,1)\rangle, \qquad \ker(R_{-2}) = \langle(1,1)\rangle.$$

*These are* not *proportional:* $(-1,1)$ *and* $(1,1)$ *span* $\mathbb{Q}^2$.

*Proof.* $R_{+2}^2 = 0$ since $R_{+2}$ is rank-1 with zero trace. Kernel of $\left(\begin{smallmatrix} -1 & -1 \\ 1 & 1 \end{smallmatrix}\right)$ is $\langle(-1,1)\rangle$. Similarly for $R_{-2}$. □

## 5.3 The residue at infinity

**Proposition 5.4.** *$R_\infty$ is semisimple with eigenvalues $\pm(2k+1)/8$. The local monodromy at infinity has eigenvalues $e^{\pm\pi i(2k+1)/4}$.*

*Proof.* $R_\infty = (c_k/2)\,\mathrm{diag}(1,-1)$, so its eigenvalues are $\pm c_k/2 = \pm(2k+1)/8$. The formal monodromy is $T_\infty = \exp(2\pi i\,R_\infty) = \mathrm{diag}(e^{\pi i(2k+1)/4},\, e^{-\pi i(2k+1)/4})$. □

## 5.4 Irreducibility and the differential Galois group

**Theorem 5.5.** *For each block $W_k$, the differential Galois group is $G_{\mathrm{gal}}(W_k) = \mathrm{SL}_2$.*

*Proof.* A rank-2 connection on $\mathbb{P}^1 \setminus \{-2,2,\infty\}$ with $\det(M_k) = c_k^2 \cdot \frac{-t^2/4+1}{(t^2-4)^2}$ has traceless residues (verified: $\mathrm{tr}(R_{\pm 2}) = \mathrm{tr}(R_\infty) = 0$), so $G_{\mathrm{gal}} \subset \mathrm{SL}_2$.

For equality, we show the connection is irreducible. A rank-1 sub-connection (invariant line) would be invariant under all local monodromies, hence lie in $\ker(R_{+2}) \cap \ker(R_{-2})$. But $\ker(R_{+2}) = \langle(-1,1)\rangle$ and $\ker(R_{-2}) = \langle(1,1)\rangle$ are non-proportional (Proposition 5.3), so no common invariant line exists.

An irreducible traceless rank-2 connection has $G_{\mathrm{gal}} = \mathrm{SL}_2$ by the Kolchin–Kovacic classification (see [17], Theorem 1.31). □

### 5.5 The exotic line bundle

The exotic connection on $\bigwedge^4(V_+)$ is determined by the trace $\tau_+(t) = M_{00} + M_{22} + M_{44} + M_{66} = 0$.

A zero trace means the connection on the line bundle $\bigwedge^4(V_+)$ is trivial: $\nabla(\omega_0 \wedge \omega_2 \wedge \omega_4 \wedge \omega_6) = 0$. The flat section is a nonzero constant—an algebraic function.

The representation theory of $\mathrm{SL}_2$ converts this per-block computation into a statement about the exotic line bundle. The key observation is elementary: $\det(\mathrm{SL}_2) = 1$.

**Corollary 5.6.** *The differential Galois group on the exotic line bundle* $\bigwedge^4(V_+) \cong \bigwedge^2(W_0) \otimes \bigwedge^2(W_2)$ *is trivial:* $G_{\mathrm{gal}}(\bigwedge^4 V_+) = \{1\}$.

*Proof.* Since $G_{\mathrm{gal}}(W_k) = \mathrm{SL}_2$ (Theorem 5.5), the induced action on $\bigwedge^2(W_k)$ is via the determinant character $\det\colon \mathrm{SL}_2 \to \mathbb{G}_{\mathrm{m}}$. But $\det(g) = 1$ for all $g \in \mathrm{SL}_2$ by definition. Hence $G_{\mathrm{gal}}$ acts trivially on each factor $\bigwedge^2(W_k)$, and therefore trivially on their tensor product.

Equivalently: the trace $\tau_+(t) = 0$ means the connection on $\bigwedge^4(V_+)$ has zero connection matrix, so every section is flat. The exotic Weil class is a *global flat section*—a constant, rational class across the entire family. □

*Remark* 5.7. The fact that $G_{\mathrm{gal}}(W_k) = \mathrm{SL}_2$ is the *strongest possible* Galois group for a traceless rank-2 connection, yet it produces the *weakest possible* Galois group on the top exterior power. This is not a paradox: irreducibility of the standard representation implies triviality of the determinant representation. The exotic class is flat *because* the monodromy is large, not despite it.

## 6 The CRM Squeeze

### 6.1 The classical boundary

The Hodge decomposition of $H^4(\mathrm{Jac}(C_t), \mathbb{C})$ produces exotic $(2,2)$-classes that exist by purely classical Hodge theory. Their existence requires:

- The Hodge decomposition (analytic, uses completeness of $\mathbb{R}$).
- The $\mathbb{Q}(i)$-action on $H^1$ (algebraic, defined over $\mathbb{Q}$).
- The cup-product structure identifying $\bigwedge^4(V_+)$ as a Hodge class space (algebraic).

### 6.2 The constructive bridge

The Griffiths reduction (Protocol 2.2) is a finite algebraic procedure over $\mathbb{Q}(t)$—no analysis required. The monodromy residues are rational numbers, the kernel computation is linear algebra over $\mathbb{Q}$, and the irreducibility criterion (non-proportional kernels) is a decidable algebraic condition.

Specifically:

1. The Bézout identity in $\mathbb{Q}(t)[x]$ is constructive (Euclidean algorithm).
2. Each matrix entry $M_{jk} \in \mathbb{Q}(t)$ is computed by finite polynomial arithmetic.
3. The Griffiths identity (1) is verified by the `ring` tactic in Lean.
4. The residues are evaluated at $t = \pm 2$ by rational arithmetic.
5. Non-proportionality of kernel vectors is checked by a $2 \times 2$ determinant.

### 6.3 The squeeze theorem

**Theorem 6.1** (CRMLint Squeeze)**.**

CLASS $\vdash$ *There exist exotic Hodge classes on* $\mathrm{Jac}(C_t)$ *for generic* $t$.

BISH $\vdash$ *Griffiths reduction* + $\mathrm{SL}_2$ *monodromy* $\Rightarrow \det = 1$
$\Rightarrow$ *exotic class is a global flat section.*

*The classical prediction (exotic Hodge classes exist) is* confirmed *by the constructive computation: the* BISH*-computable monodromy has trivial determinant, so the exotic class is monodromy-invariant. By Deligne's Theorem of the Fixed Part, it is an absolute Hodge class on every fiber.*

*Proof.* The upper bound is standard Hodge theory. For the lower bound: given $M$ with entries in $\mathbb{Q}(t)$ (Theorem 4.1), compute residues (Proposition 5.1), verify nilpotency (Proposition 5.3), extract kernels, check non-proportionality. All steps are BISH-valid (finite polynomial and linear algebra over $\mathbb{Q}$). The conclusion $G_{\mathrm{gal}}(W_k) = \mathrm{SL}_2$ combined with $\det(\mathrm{SL}_2) = 1$ gives $G_{\mathrm{gal}}(\bigwedge^4 V_+) = \{1\}$ (Corollary 5.6).

The Kolchin–Kovacic classification (irreducible + traceless $\Rightarrow \mathrm{SL}_2$) is used as a black-box lemma. Its proof is classical, but its *application* here requires only checking the (decidable) hypotheses. The CRM audit records this dependency. □

## 7 The Degree Trap Revisited

Papers 80–83 solved the "degree trap" for genus-1 families: unbounded-degree polynomial identities are replaced by the finitely-many matrix entries of the Gauss–Manin connection. Paper 84 shows the same method works at genus 4, with the additional structure coming from the $\mathbb{Q}(i)$-eigenspace decomposition.

| Paper | Family | Genus | Blocks |
|---|---|---|---|
| 80 | Legendre $y^2 = x(x-1)(x-t)$ | 1 | $1 \times (2 \times 2)$ |
| 81–83 | Legendre $E_t \times E_t$ | 1 | Kronecker sum |
| 84 | $y^2 = x^9 - tx^5 + x$ | 4 | $4 \times (2 \times 2)$ |

The genus-4 computation is qualitatively harder (degree-9 Bézout in $\mathbb{Q}(t)[x]$), but the block decomposition via $\tau$ keeps each block at rank 2, where the Kolchin–Kovacic classification applies. The connection matrix entries are rational functions of degree 1 over 2 (not degree 5 over 2 as in the erroneous v1), confirming Deligne regularity.

The key new ingredient is the coboundary correction $A_k = B_k - h'_k$. At genus 1 (Paper 80), the polynomial $h_k$ has low degree and $h'_k$ is negligible after reduction. At genus 4, $h_k$ has degree up to 8 (comparable to $\deg f = 9$), and omitting $h'_k$ introduces spurious polynomial growth—exactly the v1 bug.

# 8 CRM Audit

| Item | Status |
|---|---|
| CRM classification | BISH for Griffiths reduction, monodromy analysis, and det = 1 triviality. CLASS for Hodge class existence. |
| Omniscience gap | CLASS \ BISH: the Hodge decomposition producing exotic classes. The BISH computation *confirms* the CLASS prediction (trivial exotic monodromy). |
| Lean axioms | Zero. The main theorem uses no axioms. |
| Classical input | Hodge decomposition (upper bound); Kolchin–Kovacic classification (applied as black-box lemma). |
| Constructive content | Explicit $8 \times 8$ matrix, explicit residues, explicit kernel vectors, decidable non-proportionality check. |
| Trust boundary | CAS oracle (SymPy) for Bézout computation. Verified by `ring` in Lean (not `rfl` on strings). |

## 8.1 Logical dependencies

| Result | Level | Dependencies |
|---|---|---|
| Griffiths identity (1) | BISH | Polynomial arithmetic over $\mathbb{Q}(t)$ |
| Block decomposition (Thm 1.1) | BISH | Griffiths identity + cohomological reduction |
| $\mathrm{SL}_2$ monodromy (Thm 1.2) | BISH | Residues + kernel computation + Kolchin–Kovacic |
| Exotic flatness (Cor 5.6) | BISH | $\mathrm{SL}_2$ monodromy + $\det(\mathrm{SL}_2) = 1$ |
| CRM squeeze (Thm 1.3) | CLASS/BISH | Hodge decomposition (CLASS) + flatness (BISH) |

# 9 Formal Verification

## 9.1 File structure

The Lean 4 bundle `P84_ExoticTrace/` verifies the algebraic core of the paper:

| File | Content |
|---|---|
| `TraceData.lean` | CAS-emitted Griffiths identities, connection matrix entries, verified by `ring` |
| `Paper84_ExoticTrace.lean` | Squeeze theorem statement, axiom inventory, `#print axioms` |

### 9.2 Verification strategy

Unlike v1 (which used `rfl` on string constants), v2 verifies the algebraic content using the `ring` tactic:

1. **Griffiths identity.** For each $k = 0, \ldots, 7$: the polynomial identity $N_k(x,t) + h_k(x,t) \cdot f'(x,t) = 2\,B_k(x,t) \cdot f(x,t)$ is verified by `ring`.
2. **Connection entries.** Each nonzero $M_{jk}$ is expressed as a cross-multiplied polynomial identity (clearing denominators) and verified by `ring`.
3. **Symplectic trace.** $\sum_{k=0}^{7} M_{kk} = 0$ after clearing common denominator, verified by `ring`.
4. **Block structure.** Off-block entries verified to vanish.

### 9.3 Axiom inventory

```
#print axioms exotic_trace_squeeze
-- 'exotic_trace_squeeze' depends on axioms: []
```

Listing 1: Expected axiom output

The squeeze theorem depends on no axioms. The `ring` tactic operates in $\mathbb{Z}[x,t]$ (commutative ring arithmetic), which is axiom-free in Lean 4/Mathlib.

### 9.4 The trust boundary

| | **v1** | **v2** | **Verification** |
|---|---|---|---|
| Bézout identity | CAS | CAS | `ring` |
| $h_k$ reduction | CAS | CAS | `ring` |
| Coboundary $A_k = B_k - h'_k$ | *missing* | CAS | `ring` |
| Matrix entries | CAS | CAS | `ring` |
| Residues | CAS | CAS | `norm_num` |
| Kernel non-proportionality | — | CAS | `decide` |

The v1 trust boundary failure was *precisely* at the coboundary correction: the one step omitted from the CAS was the one step that the Lean verification (using `rfl` on CAS strings) could not catch. The v2 strategy—`ring` on polynomial identities—would have caught the bug, since the Griffiths identity (1) involves $B_k$, not $A_k$, and $A_k = B_k - h'_k$ is separately verified.

### 9.5 Reproducibility

**Toolchain:** `leanprover/lean4:v4.29.0-rc2`. **Dependency:** Mathlib (fetched automatically by `lake`).

```
cd P84_ExoticTrace && lake build
# Expected: zero errors, zero sorry
python3 ../solve_exotic_trace_v4.py
# Expected: tr(M) = 0, all identities verified
```

Listing 2: Build and verify

**Zenodo:** https://doi.org/10.5281/zenodo.18802819

# 10 Discussion

## 10.1 The function-field pipeline at genus 4

The pipeline developed in Papers 80–83 (Griffiths reduction → flat section analysis → Kovacic's algorithm → Picard number) was designed for genus-1 families. Paper 84 demonstrates that the first two steps (Griffiths reduction and monodromy analysis) extend to genus 4 with the help of the $\mathbb{Q}(i)$-eigenspace decomposition.

The key simplification is the block structure: the order-8 automorphism $\tau$ decomposes the $8 \times 8$ connection into four $2 \times 2$ blocks, each amenable to the rank-2 classification from Paper 82.

## 10.2 What the positive result means

The result $G_{\mathrm{gal}}(W_k) = \mathrm{SL}_2$ is the strongest possible answer for a traceless rank-2 connection—yet it produces the weakest possible Galois group on the exotic line bundle. This means:

- The exotic Weil class is a global flat section ($\tau_+ = 0$).
- The monodromy representation on $\bigwedge^4(V_+)$ is trivial.
- By Deligne's Theorem of the Fixed Part, the exotic class is absolute Hodge on every fiber.

The gap to the full Hodge Conjecture is the step from "absolute Hodge" to "algebraic." For abelian varieties, this is expected to hold (and is known for CM abelian varieties by work of Deligne and others [4]), but the general statement remains open.

## 10.3 Comparison: v1 vs v2

| | **v1 (wrong)** | **v2 (correct)** |
|---|---|---|
| Algorithm | $B_k$ only | $A_k = B_k - h'_k$ |
| Trace $\tau_+(t)$ | $\frac{-250t^5+\cdots}{81(t^2-4)}$ | 0 |
| Regularity | violated | satisfied |
| Singularity type | irregular | regular |
| $G_{\mathrm{gal}}$ per block | $\mathbb{G}_{\mathrm{m}}$ (exponential) | $\mathrm{SL}_2$ (monodromy) |
| $G_{\mathrm{gal}}(\bigwedge^4 V_+)$ | $\mathbb{G}_{\mathrm{m}}$ (wrong) | $\{1\}$ (det = 1) |
| Conclusion | no exotic cycles (wrong) | exotic class is flat |
| Lean verification | `rfl` on strings | `ring` on polys |
| Automorphism | $\sigma^2$ (wrong) | $\tau$ (order 8) |

The conclusion *flips* between versions. The v1 claim of "no exotic cycles" was wrong in both mechanism and conclusion: the irregular singularity does not exist, and the exotic class is in fact a global flat section. The corrected computation ($G_{\mathrm{gal}} = \mathrm{SL}_2$ per block) gives $\det(\mathrm{SL}_2) = 1$, so the action on $\bigwedge^4(V_+)$ is trivial. The exotic Weil class is monodromy-invariant—a positive result.

## 10.4 Extension to other families

The method applies to any genus-4 family with $\mathbb{Q}(i)$-action where:

1. The Bézout computation in $\mathbb{Q}(t)[x]$ terminates (always, for polynomial $f$).
2. The eigenspace decomposition produces rank-2 blocks.
3. The monodromy residues can be computed explicitly.

The result $\tau_+ = 0$ (trivial exotic Galois group) holds for the family $y^2 = x^9 - tx^5 + x$ studied here. Whether this vanishing is specific to this family or universal for genus-4 $\mathbb{Q}(i)$-families is addressed in Paper 85.

### 10.5 The Tannakian perspective

The Gauss–Manin connection on $H^1_{\mathrm{dR}}(C_t/\mathbb{Q}(t))$ is an object in the Tannakian category of algebraic connections over $\mathbb{Q}(t)$. The differential Galois group $G_{\mathrm{gal}}$ is the Tannakian fundamental group.

The block decomposition $M = \bigoplus_{k=0}^{3} M_k$ corresponds to a direct sum decomposition in the Tannakian category. Each block $M_k$ with $G_{\mathrm{gal}}(M_k) = \mathrm{SL}_2$ is a simple object.

The exotic connection on $\bigwedge^4(V_+)$ is a sub-quotient of the exterior power. Since the individual blocks are $\mathrm{SL}_2$-connections, and $\mathrm{SL}_2$ representations are rigid, the exotic flat section space is determined by representation-theoretic constraints: the determinant character $\det\colon \mathrm{SL}_2 \to \mathbb{G}_{\mathrm{m}}$ is trivial, so $\bigwedge^2(W_k)$ is the trivial representation for each $k$. The exotic connection on $\bigwedge^4(V_+) \cong \bigwedge^2(W_0) \otimes \bigwedge^2(W_2)$ is therefore a tensor product of trivial representations—itself trivial.

### 10.6 The CRM pipeline across genera

| Paper | Family | Genus | $G_{\mathrm{gal}}$ | Result |
|---|---|---|---|---|
| 80 | Legendre | 1 | $\mathrm{SL}_2$ | Connection computed |
| 81 | $E_t \times E_t$ | 1 | $\mathrm{SL}_2$ | No extra flat sections |
| 82 | Legendre | 1 | $\mathrm{SL}_2$ | Kovacic certificate |
| 83 | $E_t \times E_t$ | 1 | — | rkNS = 3 |
| 84 | $x^9 - tx^5 + x$ | 4 | $\mathrm{SL}_2^4$ | Exotic class is flat |

## 11 Conclusion

We have computed the Gauss–Manin connection for the genus-4 family $y^2 = x^9 - tx^5 + x$ with $\mathbb{Q}(i)$-multiplication, correcting two errors in v1 (missing coboundary term, wrong automorphism). The corrected connection matrix decomposes into four $2 \times 2$ blocks via the order-8 automorphism $\tau$, each with differential Galois group $\mathrm{SL}_2$ (proven by non-proportional nilpotent kernels at the two finite singular points).

The positive result—the exotic Weil class is a global flat section—follows from the elementary representation-theoretic fact $\det(\mathrm{SL}_2) = 1$. The $\mathrm{SL}_2$ monodromy per block, far from obstructing algebraicity, *certifies* it at the level of the exotic line bundle. By Deligne's Theorem of the Fixed Part, the exotic class is absolute Hodge on every fiber.

The Lean formalization verifies all algebraic identities by the `ring` tactic, closing the CAS trust gap that produced the v1 errors. This is the seventh CRMLint squeeze and the first application of the function-field pipeline beyond genus 1.

Two directions remain open:

1. *Universality.* Is $\tau_+ = 0$ specific to this family, or universal for all genus-4 hyperelliptic families with $\mathbb{Q}(i)$-action? Paper 85 investigates this question.
2. *CAS-free pipeline.* Implement the Bézout reduction natively in Lean/Mathlib, eliminating the CAS oracle entirely.

## Acknowledgments

AI assistance was used for CAS computation (SymPy), Lean 4 formalization, and manuscript preparation. The CRM series format follows the guide [19]. The Lean 4 proof assistant and the Mathlib library [16] provide the formal verification infrastructure.

The v1 errors were identified through careful comparison with Deligne's regularity theorem and independent recomputation of the Griffiths reduction, demonstrating the value of multiple verification layers in CAS-assisted proofs.

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